

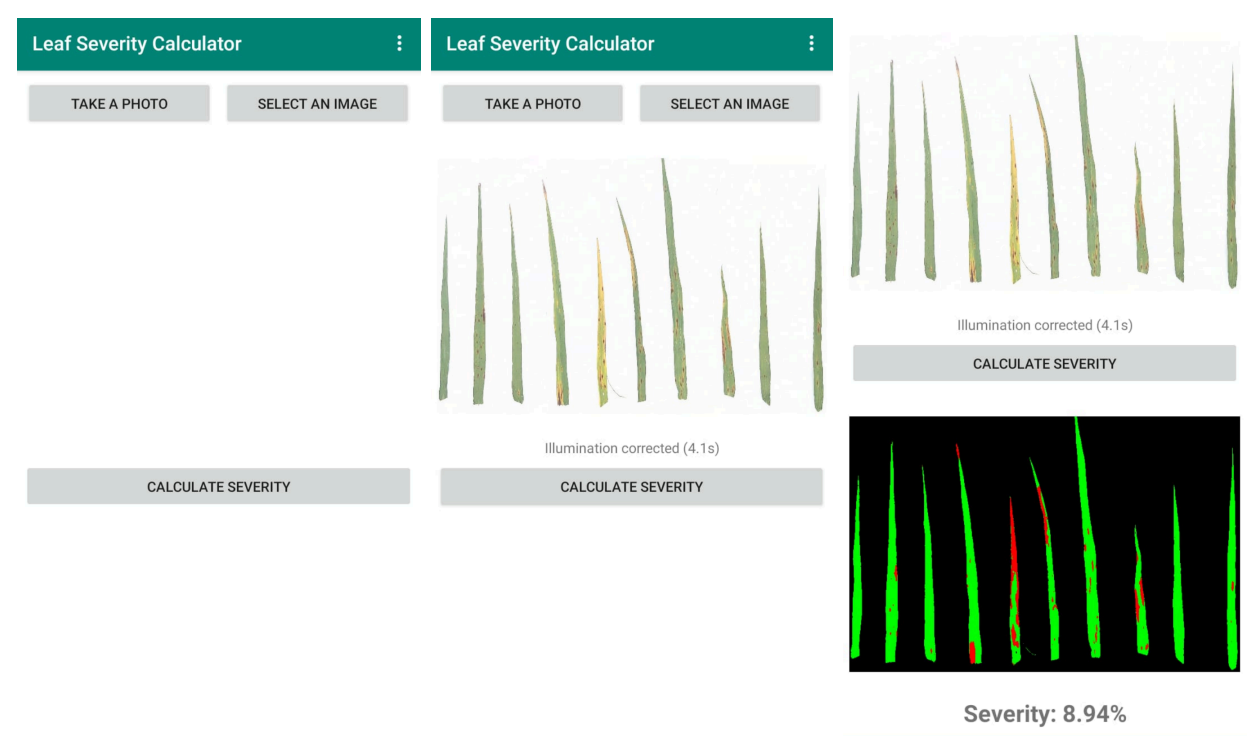
Leaf Severity Calculator - Main Branch

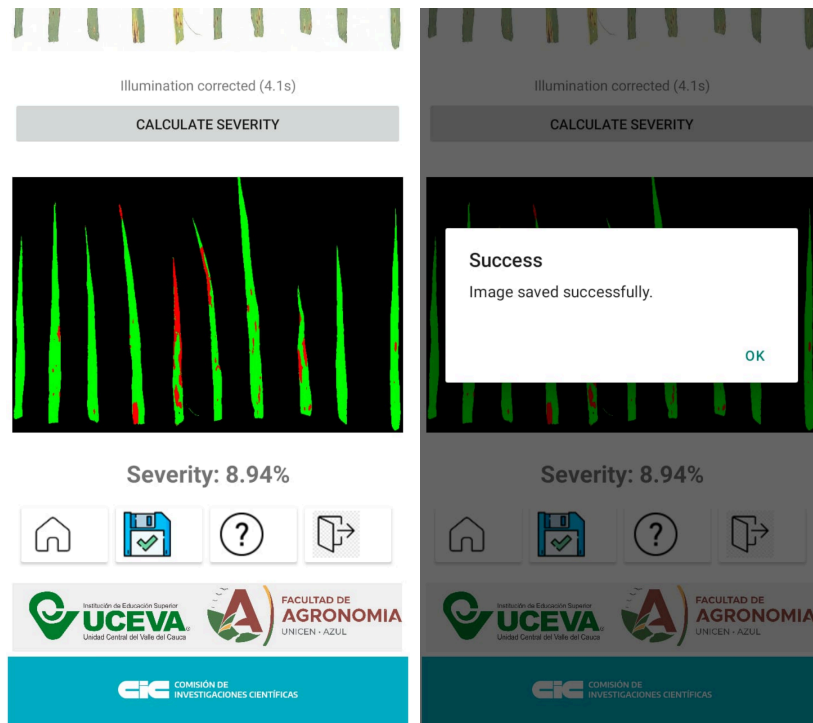
Version: 0.0.9a1 | **Stack:** Python 3.12 + Briefcase 0.3.22 | **Status:** ✔ Production Ready

Overview

A cross-platform Android application that analyzes barley leaf images to calculate disease severity. The app segments leaves into background, healthy tissue (green), and diseased tissue (red), then computes the severity percentage.

App Screenshots





Key Features

- 📷 **Capture & Upload:** Take photos with device camera or select images from gallery
- 🔬 **Threshold-Based Segmentation:** Uses fixed thresholds on blue channel (ub) and index $(green-red)/(green+red)$ (ui); these values were calibrated offline using Otsu/K-means during model setup
- 📊 **Severity Calculation:** Real-time calculation of disease percentage
- 💾 **Save Results:** Export processed images with severity metrics to device storage
- 🌐 **English USA UI:** Full English interface across dialogs, labels, and actions
- ⚡ **Modern Stack:** Python 3.12, latest PyPI dependencies

Technology Stack

Component	Version	Source
Python	3.12	System
Briefcase	0.3.22	PyPI
toga-android	~0.4.5	PyPI

Component	Version	Source
NumPy	Latest	PyPI
Pillow	Latest	PyPI
numpy-rolling-ball	>=1,<2	PyPI
tatogalib	Latest	Local wheel

Branch Processing Notes (Main vs OpenCV-based)

Both branches keep the same functional workflow:

- Camera/gallery input
- Illumination correction before severity
- Aspect-ratio-preserving resize (fit inside 800x600)
- Threshold segmentation and severity display

The implementation intentionally differs:

- **main**
 - Illumination correction: `numpy-rolling-ball`
 - Processing style: NumPy-first pipeline (modern stack)
- **OpenCV-based**
 - Illumination correction: `opencv-rolling-ball` (`cv2_rolling_ball`)
 - Processing style: OpenCV for decode/resize/encode + NumPy masks for severity math (legacy stack)

This split reflects compatibility and memory constraints across modern vs legacy Android targets.

Project Structure

```
leafseveritycalculator/
├─ src/leafseveritycalculator/
│   └─ __init__.py
```

```

|   ├── app.py                # Main application logic
|   ├── __main__.py
|   └── resources/           # App resources
├── android/
|   └── app_template/        # Android-specific templates
|       └── src/main/
|           ├── AndroidManifest.xml
|           └── res/         # Drawable, layout, strings, etc.
├── tests/                   # Test suite
├── wheels/                  # Universal wheels (tatogalib)
└── pyproject.toml           # Project configuration

```

Why there is a second README/LICENSE inside

`leafseveritycalculator/`

That folder is the Briefcase app project root used for packaging. Keeping `README.rst` and `LICENSE` there is expected, while root-level docs are repository-level documentation.

- Root docs/files: repository documentation for contributors/GitHub
- `leafseveritycalculator/README.rst` and `leafseveritycalculator/LICENSE`: app packaging metadata/docs

Getting Started

Prerequisites

- **Python 3.12** (Windows, macOS, or Linux)
- **Android SDK** (API level 24+)
- **Android NDK**
- **Java Development Kit (JDK)** 11+
- **Briefcase** 0.3.22

Installation & Setup

1. **Clone the repository:**

```
git clone https://github.com/alencina-faa/Leaf-Severity-Calculator_android
cd Leaf-Severity-Calculator_android
git checkout main # Ensure you're on the main branch
```

2. Create a virtual environment:

```
python3.12 -m venv venv
source venv/bin/activate # On Windows: venv\Scripts\activate
```

3. Install dependencies:

```
pip install --upgrade pip setuptools wheel
pip install briefcase==0.3.22
```

Building for Android

Development/Debug Build

```
cd leafseveritycalculator

# Initial setup (one-time)
python -m briefcase create android --no-input

# Build debug APK
python -m briefcase build android

# Package debug APK
python -m briefcase package android
```

Output: dist/LeafSeverityCalculator-0.0.9a1.apk

Release Build

```
cd leafseveritycalculator
```

```
# Update dependencies
python -m briefcase update android -r

# Package release AAB (for Google Play)
python -m briefcase package android
```

Output: `dist/LeafSeverityCalculator-0.0.9a1.aab`

Running on Device

1. **Transfer APK to Android device** (or deploy directly if connected):

```
adb install dist/LeafSeverityCalculator-0.0.9a1.apk
```

2. **Grant permissions:**

- Camera access (prompted on first use)
- Storage access (for saving results)

3. **Launch the app** from device app drawer

Key Dependencies Explained

toga-android

BeeWare's native Android UI toolkit. Provides cross-platform widgets that render as native Android components.

numpy & Pillow

Used for image processing:

- NumPy: Array operations, channel extraction, mathematical operations
- Pillow: Image I/O, resizing, format conversion

numpy-rolling-ball

Background illumination correction using rolling-ball algorithm (subtracts background light to normalize images).

tatogalib

URI file browser and Android file system integration (allows selecting images from device gallery).

Development Workflow

Running Tests

```
cd leafseveritycalculator
python -m pytest tests/
```

Code Structure

Main Application (`src/leafseveritycalculator/app.py`):

- `LeafSeverityCalculator` : Main app class
- `startup()` : Initialize UI and handlers
- `take_photo()` : Camera integration
- `open_image()` : Gallery file picker
- `extract_background_color()` : Illumination correction
- `process_image()` : Image segmentation and severity calculation
- `save_image()` : Save results to device storage

UI Components

- **Buttons**: Take Photo, Select Image, Calculate Severity
- **Image Viewers**: Input photo, processed result
- **Progress Label**: Status updates during processing
- **Result Label**: Displays severity percentage
- **Icon Buttons**: Home, Save, Help, Exit
- **Institutional Logos**: UCEVA, FAA, CIC

Android Permissions

Defined in `android/app_template/src/main/AndroidManifest.xml` :

```
<uses-permission android:name="android.permission.CAMERA" />
<uses-permission android:name="android.permission.READ_EXTERNAL_STORAGE" />
<uses-permission android:name="android.permission.WRITE_EXTERNAL_STORAGE" />
```

Building Troubleshooting

Issue: "Version number not valid"

Solution: Ensure version in `pyproject.toml` is PEP 440 compliant (e.g., `0.0.9a1` , not `0.0.9a`)

Issue: Module not found errors

Solution: Ensure all dependencies are installed via pip:

```
pip install briefcase toga-android numpy pillow numpy-rolling-ball tatogalib
```

Issue: Android SDK/NDK not found

Solution: Set environment variables:

```
export ANDROID_HOME=$HOME/Android/Sdk
export ANDROID_SDK_ROOT=$ANDROID_HOME
export PATH=$PATH:$ANDROID_HOME/tools:$ANDROID_HOME/platform-tools
```

Deployment to Google Play

1. **Prepare AAB** (already done):


```
python -m briefcase package android
```

2. **Sign release** (configured in `pyproject.toml`)

3. **Upload to Play Console:**

- Navigate to [Google Play Console](#)
- Select LeafSeverityCalculator app
- Go to **Release** → **Production**
- Upload `dist/LeafSeverityCalculator-0.0.9a1.aab`
- Add release notes
- Review and publish

Environment Variables

For CI/CD workflows:

ANDROID_HOME	# Path to Android SDK
ANDROID_SDK_ROOT	# Alternative Android SDK path
ANDROID_NDK_ROOT	# Path to Android NDK
JAVA_HOME	# Path to JDK installation

Version Information

- **Current:** 0.0.9a1 (Alpha Release)
- **Branch:** main (Production-ready modern stack)
- **Python:** 3.12.x
- **Briefcase:** 0.3.22
- **Last Updated:** April 24, 2026

Related Branches

- **OpenCV-based:** Legacy Python 3.8 stack for broader device compatibility
 - Better support for older Android devices
 - Uses pre-built cp38 wheels

- See [OpenCV-based](#) branch README

Contributing

1. Create feature branch from `main`
2. Make changes and test locally
3. Update version in `pyproject.toml` (PEP 440 format)
4. Commit with descriptive messages
5. Create pull request with details

License

See [LICENSE](#) file.

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Additional Resources

- [BeeWare Briefcase Documentation](#)
 - [Toga UI Toolkit](#)
 - [Android Developer Guide](#)
 - [Python for Android](#)
 - [Chaquopy \(Python-Android build system\)](#)
-

Status:  Production Ready | **Last Build:** LeafSeverityCalculator-0.0.9a1.aab (43.21 MB)